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| Systematic Review / Sistematik Derleme |

Pedagogical Impacts of Block-Based Artificial Intelligence Applications: A Systematic Review

Blok Tabanlı Yapay Zekâ Uygulamalarının Pedagojik Etkileri: Sistematik Bir Derleme[1]

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Keywords

1. Artificial intelligence education
2. Block-based programming
3. K-12 education
4. Pedagogical impact
5. Computational thinking

Anahtar Kelimeler

1. Yapay zekâ eğitimi
2. Blok tabanlı programlama
3. K-12 eğitimi
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Abstract

Purpose: The purpose of this systematic review is to methodically investigate the pedagogical effects of visual and block-based programming tools used in K–12 artificial intelligence (AI) education.

Method: A systematic review was conducted following the PRISMA 2020 guidelines, analyzing 91 articles published between 2019 and 2024 that met the inclusion criteria. The studies were evaluated considering the block-based tools used at various educational levels (elementary, middle, high school, K–12), the AI topics taught, their pedagogical effects, and recommendations for future research.

Results: The results show that visual and block-based programming tools are useful for improving students' comprehension of AI principles, problem-solving ability, computational thinking skills, and ethical awareness. The use of these technologies varies across different countries and educational levels; specifically, more advanced applications are observed in OECD member countries due to their superior infrastructure and resources.

Highlights: To increase the efficacy of AI education, the study emphasizes the need to strengthen teacher preparation, integrate ethical and social concerns into the curriculum, and develop real-world learning environments. It concludes that block-based and visual programming tools are essential components of K–12 AI instruction, holding significant potential to enhance students' cognitive, affective, and psychomotor skills.

Öz

Amaç: Bu çalışmanın amacı, K–12 düzeyinde yapay zekâ (YZ) öğretiminde kullanılan görsel ve blok tabanlı programlama araçlarının pedagojik etkilerini sistematik biçimde incelemektir.

Yöntem: PRISMA 2020 standartlarına uygun olarak yürütülen sistematik derleme kapsamında, 2019–2024 yılları arasında yayımlanan ve belirlenen ölçütleri karşılayan 91 makale analiz edilmiştir. İnceleme sürecinde farklı eğitim kademelerinde (ilkokul, ortaokul, lise, K–12) kullanılan araçlar, YZ ile öğretilen konular, pedagojik etkiler ve gelecekteki araştırmalara yönelik öneriler dikkate alınmıştır.

Bulgular: Bulgular, görsel ve blok tabanlı programlama araçlarının öğrencilerin YZ kavramlarına yönelik anlayışlarını, problem çözme becerilerini, hesaplamalı düşünme yetilerini ve etik farkındalıklarını geliştirmede etkili olduğunu göstermektedir. Bu teknolojilerin kullanım düzeyi ülkeler arasında ve eğitim kademelerine göre değişiklik göstermekte; özellikle OECD üyesi ülkelerde daha gelişmiş altyapı ve kaynaklar nedeniyle daha ileri düzeyde kullanılmaktadır.

Önemli Vurgular: Çalışma, YZ eğitimini daha etkili hâle getirmek için öğretmen eğitimlerinin güçlendirilmesi, müfredata etik ve sosyal boyutların dâhil edilmesi ve uygulamaya dayalı öğrenme ortamlarının geliştirilmesi gerektiğini vurgulamaktadır. Sonuç olarak, blok tabanlı ve görsel programlama araçlarının K–12 YZ öğretiminin önemli bileşenleri olduğu ve öğrencilerin bilişsel, duyuşsal ve psikomotor becerilerini geliştirmeye katkı sağlayabileceği belirtilmektedir.

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INTRODUCTION

Artificial intelligence (AI) was defined in 1956 as "the science and engineering of making intelligent machines" (McCarthy, 1987). Since that early conceptualization, the world has undergone a profound technological shift driven by the rapid and widespread adoption of AI systems. Once rooted primarily in computer science, AI now permeates domains such as engineering, business, healthcare, and the arts, influencing nearly every aspect of daily life. Everyday technologies—including voice assistants, smartphones, cloud-based services, and smart home devices—demonstrate how AI enhances user experience, boosts productivity, and simplifies a wide range of tasks (Tedre et al., 2021; Van Brummelen et al., 2021). As AI continues to expand, individuals increasingly need new competencies to function effectively in the information age, making AI literacy a fundamental requirement (Estevez et al., 2019).

The importance of equipping students for future, automation-driven societies is emphasized in the OECD report Artificial Intelligence in Education: Promises and Implications for Teaching and Learning (Vincent-Lancrin & van der Vlies, 2020). Alongside established literacies such as digital, data, and information literacy, AI literacy is now considered one of the core technological literacies of the twenty-first century (Long & Magerko, 2020). Consequently, scholars argue that an AI education ecosystem spanning all grade levels—not only higher education—must be developed to prepare students for an AI-augmented world (Basu, 2019; Kandlhofer et al., 2016). In response, several countries, including the United States, the United Kingdom, Finland, China, Australia, and South Korea, have initiated national strategies, curriculum reforms, and teacher training programs to integrate AI into K–12 education (UNESCO, 2019).

Türkiye exhibits a parallel trajectory. The Ministry of National Education (MoNE) has begun embedding AI-related competencies into the national curriculum, particularly within the framework of the 2023 Education Vision and the Presidential Strategic Plan. One notable initiative is the Project on Artificial Intelligence in Education for Increasing Opportunities (FEYZA), which aims to introduce AI education to students from an early age and strengthen national capacity in digital transformation (MoNE, 2023). These reforms indicate a clear national commitment to building AI literacy and ensuring that Türkiye's education system adapts to emerging technological demands.

At the classroom level, the primary goal of AI education is to develop students' critical thinking, problem-solving, and algorithmic reasoning skills (Park & Shin, 2021). Block-based programming environments, such as Scratch, play a significant role in achieving this goal by providing a visual and intuitive interface for exploring computational and AI concepts. Extensions such as Tooee and DeepScratch further enhance these platforms by enabling learners to build more advanced AI-driven applications (Park & Shin, 2022a; Alturayeif et al., 2020). Beyond technical skills, scholars emphasize the necessity of cultivating ethical awareness. Understanding algorithmic bias, fairness, and responsible AI use helps students develop informed and critical perspectives on technology (Hsu et al., 2021). Educational initiatives such as How to Train Your Robot illustrate how middle school learners can meaningfully engage with ethical dilemmas while developing foundational AI concepts (Williams et al., 2021).

Despite rapidly increasing global interest, research on AI education—particularly studies involving block-based AI tools—remains fragmented. Existing studies vary widely in terms of learning outcomes, age groups, instructional approaches, and methodological rigor, making it difficult to identify broader patterns or evidence-based practices. Additionally, only a limited number of reviews synthesize findings specifically for K–12 contexts, leaving gaps in understanding regarding the cognitive, affective, and ethical outcomes associated with block-based AI learning environments.

To address these gaps, this study presents a systematic review of empirical research on block-based AI education in K–12 settings. By analyzing studies published between 2019 and 2024, the review aims to synthesize evidence on instructional approaches, learning outcomes, assessment practices, and methodological patterns.

In line with the gaps identified in the literature, this study aims to synthesize empirical research on block-based AI education in K–12 contexts by addressing the following research questions:

1. Which block-based and visual programming tools are used in K–12 AI education, and how are these tools distributed across different grade levels?
2. What pedagogical outcomes—cognitive, affective, and ethical—are associated with block-based AI learning environments?
3. What methodological trends, limitations, and future research directions are reported in existing studies on block-based AI education?
4. How is block-based AI education geographically distributed across countries, and to what extent do factors such as OECD membership relate to these implementations?

LITERATURE REVIEW

The rapid advancement of digital transformation has elevated artificial intelligence (AI) education to a central position within K–12 learning environments. AI supports students in developing future-oriented skills and preparing for an increasingly automated workforce (Williams et al., 2019; Zhou et al., 2024). Central to its value is AI's capacity to provide personalized learning

opportunities tailored to individual learning styles and paces (Park & Shin, 2022a; Hsu et al., 2022). Beyond academic achievement, AI education has been associated with positive impacts on students' cognitive development (Kong et al., 2022). Yet the abstract nature of AI concepts poses pedagogical challenges, especially for younger learners, making it necessary to design instructional materials in alignment with children's cognitive development (Park & Shin, 2022a; Williams et al., 2019). In response to these challenges, block-based and visual programming tools have gained prominence due to their effectiveness in concretizing otherwise complex or invisible AI mechanisms (Emerson et al., 2020).

AI Education in Early Childhood and Elementary School

Research on early AI education consistently highlights the importance of tangible, interactive, and embodied learning environments. Vartiainen et al. (2020) demonstrated that children aged 3–9 can meaningfully engage with foundational AI and machine learning (ML) ideas through rapid cognitive associations and physical actions when using tools such as Google's Teachable Machine. Similarly, Kim and Kwon (2024) found that physical computing environments support younger students in forming more coherent mental models of AI, thereby positively shaping their perceptions of the technology. Interdisciplinary curriculum designs have also been shown to enrich early AI learning by helping children make connections between AI and subjects such as biology, physics, and sociology (Sabuncuoğlu, 2020).

At these levels, visual and block-based programming tools serve as developmentally appropriate introductions to AI. Platforms such as Machine Learning for Kids and Teachable Machine reduce cognitive load by representing abstract operations through manipulable blocks, thereby allowing even novice learners to explore foundational concepts of classification, pattern recognition, and training datasets (Yim & Su, 2025). These environments support the formation of early computational thinking skills, which serve as critical building blocks for later engagement with more advanced AI systems (Casal-Otero et al., 2023).

AI Education in Middle School

In middle school contexts, block-based programming becomes a key medium through which students learn algorithmic reasoning, problem-solving, and AI fundamentals. Visual programming environments such as Scratch, Blockly, and Snap! have been widely adopted for AI instruction because they eliminate the complexities of syntax inherent in text-based coding (Kazemitabaar et al., 2023; Williams et al., 2021; Zhou et al., 2024). Evidence shows that these tools help students understand complex algorithms and conceptualize the logic behind AI models more effectively (Emerson et al., 2020). Furthermore, the use of block-based tools has been shown to enhance student motivation and foster active engagement (Kazemitabaar et al., 2023).

Maker-centered and project-based learning approaches further support conceptual understanding at this level. Ng et al. (2023) reported that middle school students who worked with AI-enabled recycling bins demonstrated increases in motivation, collaboration, and AI literacy. Project-based environments that connect AI to students' personal experiences or everyday phenomena have also been found to improve conceptual understanding and promote higher-order cognitive abilities (Vartiainen et al., 2020; Grover, 2024). These findings highlight the value of interactive, real-world-oriented learning designs for supporting middle school students' developing computational thinking and AI reasoning.

AI Education in High School

At the secondary level, AI education typically integrates technical competencies with ethical, societal, and interdisciplinary dimensions. Estevez et al. (2019) found that a Scratch-based AI workshop improved high school students' understanding of algorithms and strengthened their computational thinking skills. Grover (2024) emphasizes that effective AI instruction at this level must balance foundational technical knowledge with critical discussions about fairness, bias, and the societal consequences of AI systems.

Digital game-based learning has also emerged as a promising approach. Alam (2022) demonstrated that game-based activities can enhance student motivation to learn complex AI concepts, while Estevez et al. (2019) showed that gamified Scratch environments create engaging conditions for learning AI fundamentals. Sabuncuoğlu (2020) similarly found that game-based experiences increase students' interest in AI and ML concepts, suggesting that such environments may serve as effective vehicles for exploring both technical and ethical facets of AI.

Block-Based AI Tools and Pedagogical Outcomes Across K–12

Across K–12 settings, block-based AI tools lower barriers to entry by reducing syntax-related cognitive load and enabling students to focus on conceptual understanding (Carrisi et al., 2025; Yim & Su, 2025). These tools foster a range of pedagogical outcomes, including improvements in computational thinking, abstraction, decomposition, and algorithmic reasoning. At the same time, block-based AI environments contribute to affective and ethical outcomes by increasing students' interest in computing, supporting collaborative learning, and strengthening ethical awareness regarding issues such as bias and transparency (Schaper et al., 2023; Williams et al., 2024).

The AI4K12 framework further reinforces the importance of block-based tools by situating AI education around five "big ideas"—perception, representation and reasoning, learning, natural interaction, and societal impact (Baldoni et al., 2024). These guidelines highlight that AI literacy must extend beyond tool proficiency to include critical thinking, problem solving, and ethical evaluation. Interdisciplinary approaches that integrate AI into mathematics, science, and social studies are consistently discussed

as contributing to students' ethical awareness and as supporting preparation for AI-related career pathways (Kaplan & Meylani, 2025). However, the literature also emphasizes that the educational potential of such approaches is contingent upon robust teacher preparation and sustained professional development, particularly with respect to pedagogical competence in AI-supported learning environments and the explicit integration of ethical considerations into AI curricula (Kaplan & Meylani, 2025).

Despite growing interest in AI and block-based programming within K–12 contexts, the existing literature remains fragmented. Prior reviews have tended to examine K–12 AI literacy broadly without focusing specifically on the pedagogical impacts of block-based AI tools (Yim & Su, 2025). Moreover, many empirical studies are small in scale, limited to individual grade levels, or centered on singular tools, making it difficult to form a holistic understanding of how block-based AI environments function across the entire K–12 spectrum (Zhou et al., 2024). Geographic disparities further complicate the landscape; for example, OECD countries tend to implement more advanced block-based AI applications due to superior infrastructure and resources.

Addressing these limitations, the present systematic review synthesizes findings from 91 studies to provide a comprehensive account of the visual and block-based tools used in K–12 AI education, their pedagogical impacts, methodological trends, and global distribution (Memari & Ruggles, 2025; Casal-Otero et al., 2023). By doing so, it responds to the need for a structured, evidence-based understanding of how block-based AI education shapes cognitive, affective, and ethical learning outcomes across diverse educational settings.

METHOD

The educational effects of block-based artificial intelligence (AI) applications at the K–12 level were examined using the systematic review methodology. The research design followed the PRISMA 2020 criteria (Page et al., 2021), a globally accepted standard for systematic reviews and meta-analyses. The review process consisted of database search, study selection, data collection, coding, thematic analysis, reliability procedures, ethical compliance, and data synthesis. This structured approach ensured an organized presentation of findings and facilitated a comprehensive analysis of the relevant literature. In this context, the selected studies were analyzed using predefined criteria.

During the data collection phase, Web of Science and Google Scholar were searched using keywords such as 'block-based tools,' 'visual programming,' 'AI education,' 'K-12,' 'artificial intelligence,' and 'machine learning.' The initial search yielded a total of 147,723 records (147,000 from Google Scholar and 723 from Web of Science). Due to the excessive volume of records retrieved from Google Scholar, a systematic data preprocessing strategy was employed to ensure methodological manageability. First, automated filters were applied to restrict results to the 2019–2024 timeframe and English language publications. Second, consistent with established systematic review protocols for high-volume databases, the screening for Google Scholar was prioritized by relevance algorithms, focusing on the first 1,000 most relevant entries after filtering, as relevance significantly diminishes beyond this threshold. Consequently, 140,630 records were excluded through these automated filtering and relevance-based truncation processes prior to the manual screening phase. The remaining 6,370 records were subsequently screened by title and abstract.

The inclusion criteria encompassed articles published in English between 2019 and 2024, studies conducted with K–12 students, and research investigating the educational effects of block-based AI applications. Both conference proceedings and peer-reviewed journal articles were considered.

The exclusion criteria were defined as studies conducted at levels other than K–12, research not focusing on the educational effects of AI, studies without accessible full texts or with incomplete data, and theoretical papers without empirical findings.

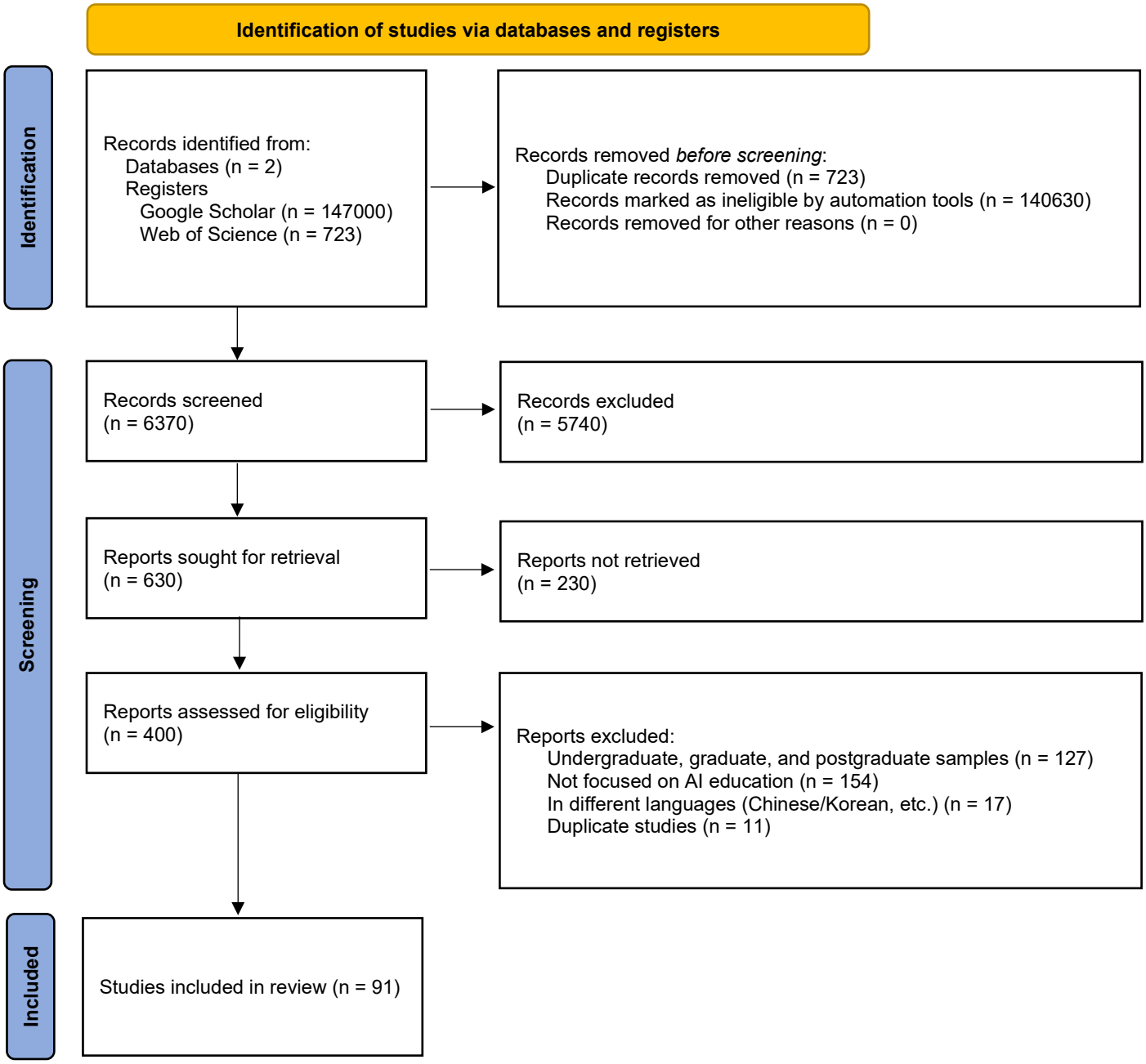


Figure 1. Flow diagram of database search and inclusion for the systematic review.

Study Selection and Inclusion

Exclusion criteria included studies that were not at the K–12 level, studies that did not focus on the educational effects of AI, research that were undertaken at levels other than K–12, and studies that had incomplete data or full texts that were not available:

1. Date of Publication: Research must have been released in 2019–2024. The usage of block-based programming tools for AI teaching got more systematic throughout this time, and it incorporates the most recent research and methods.
2. Study Type: Only empirical research was assessed. Reviews, investigations, and theoretical works that provide methodological justifications were excluded.
3. Educational Level: The research looks at how pupils are affected by block-based programming tools used in K–12 AI teaching.
4. Research Focus: The studies examine the impact of block-based or visual programming tools used to teach AI concepts on learning and investigate pedagogical or cognitive results.
5. Excluded were studies that just offered theoretical knowledge or broad suggestions and did not fit these requirements. We looked at ninety-one studies that satisfied the inclusion requirements.

A comprehensive coding schema was developed based on the data from each chosen study in order to collect data. Data were coded for each study, including dependent variables (student achievement, cognitive development, attitude changes), independent variables (block-based tools used, educational level, implementation duration), measurement techniques (tests, questionnaires, observations, etc.), sample size, and participant characteristics. The coding method was conducted by two separate researchers, which reduced the possibility of subjectivity. The inter-rater reliability (IRR) of the coding process was

assessed using Cohen's Kappa. A sample of 20% (18/91) of the included articles was double-coded independently, yielding a coefficient of 0.84. According to established benchmarks (Landis & Koch, 1977), this result indicates strong agreement between the coders.

FINDINGS

This section presents the findings obtained from the systematic analysis of 91 studies meeting the inclusion criteria. The results are organized under keywords, descriptive characteristics, pedagogical impacts, and methodological recommendations.

Analysis of Keywords Used in Studies

The common keywords and themes of the research under review are highlighted via word clouds and other data visualization techniques. The analysis shows that the most frequently used terms reflect orientations such as "AI," "Students," "Learning," "Education," "Programming," "Data," "Machine," "Children," "Design," "ML," "Knowledge," "Curriculum," "Activities," "Computing," and "Thinking." These findings suggest that AI education provides a comprehensive educational experience and that students acquire both technical and higher-order cognitive thinking skills.



Figure 2. Word cloud analysis showing the frequency of words used in the studies.

Study Coding

Full texts of the studies were tagged under headings like "author and year," "country where the study was conducted," "educational level," "used AI tool/application," "taught or addressed AI topics," "recommendations expressed in the study," and "pedagogical impacts." For easy access to the studies that were part of the coding, a hyperlink was provided to the "author and year" section.

Table 1. Descriptive information about the studies

Author and Year	OECD Membership Status	Country Where Study Was Conducted	Educational Level	Artificial Intelligence Tool/Application Used	Taught/Addressed Artificial Intelligence Topics
Akram et al., 2022	Member	USA	Middle School	Snap!	Search algorithms, knowledge representation systems, machine learning, and natural language processing
Basu, 2019	Member	USA	Middle School	Scratch and App Inventor	Block-based programming, coding skills, design processes
Broll and Grover, 2023	Member	USA	High School	NetsBlox	Classification, decision trees, optimization, adversarial examples
Chung and Shamir, 2021	Member	USA	K-12	Scratch	Machine learning, classification, robot control algorithms

Author and Year	OECD Membership Status	Country Where Study Was Conducted	Educational Level	Artificial Intelligence Tool/Application Used	Taught/Addressed Artificial Intelligence Topics
Grover et al., 2023	Member	USA	High School	NetsBlox	AI-based cybersecurity, ethical and bias issues
Grover, 2024	Member	USA	K-12	Google's Quick, Draw!, TensorFlow Playground, ML4Kids	AI teaching techniques, large language models, ethical issues
Irgens et al., 2022	Member	USA	Primary and Middle School	Scratch, Robotics, and Machine Learning Tools	Critical machine learning (ML), robot development, social and ethical orientations
Jayasuriya et al., 2024	Member	USA	Middle School	Google Teachable Machine	Instruction on visual computing, computer vision, and machine learning
Jordan et al., 2021	Member	USA	Middle School	PoseBlocks, Scratch, Teachable Machine	Body and face recognition, pose estimation, emotion recognition, AI ethical problems
Kim and Kwon, 2024	Member	USA	Primary School	Tangible Computing Tools	Reinforcement of AI knowledge, ethical awareness, coding taught with tangible tools
Kumar and Worsley, 2023	Member	USA	Middle School	Scratch, Pose Detection, IMU-based Motion Detection Tools	AI-supported sports applications, motion detection, data analysis
Kurz et al., 2024	Member	USA	Middle School	Teachable Machine	Image recognition, machine learning, science applications
Lee et al., 2021	Member	USA	Primary and Middle School	PRIMARYAI	Life sciences, AI-based problem-solving, game-based learning
Long et al., 2022	Member	USA	Primary and Middle School	Scratch	AI literacy, problem-solving, computational thinking skills
Okolo, 2024	Member	USA	Middle and High School	Generative AI Tools	AI literacy, ethical awareness, skills in working with AI were taught
Park et al., 2021	Member	USA	Primary School	PRIMARYAI	AI planning, representation, and reasoning
Park et al., 2022	Member	USA	Primary School	PRIMARYAI	AI planning, representation, and reasoning
Percival et al., 2022	Member	USA	Middle School	CryptoScratch	Symmetric and asymmetric encryption algorithms
Reddy et al., 2021	Member	USA	Middle School	Scratch, TensorFlow	K-Nearest Neighbors, text classification
Reddy et al., 2022	Member	USA	K-12	LevelUp	Supervised learning, automatic assessment
Taylor et al., 2019	Member	USA	Middle School	IntelliBlox, Unity	Block-based programming, computational thinking
Touretzky et al., 2019	Member	USA	K-12	AI4K12	The five big ideas of AI, ethics, societal impacts of AI
Touretzky et al., 2023	Member	USA	K-12	AI4K12	The five big ideas of AI, machine learning, societal impacts of AI
von Wangenheim et al., 2021	Member	USA	K-12	MIT App Inventor, Teachable Machine	Speech-based AI, AI literacy, AI ethics
Walsh et al., 2023	Member	USA	Middle and High School	AI Ethics Modules	AI ethics, algorithmic bias
Wan et al., 2020	Member	USA	K-12	SmileyCluster	Scientific discovery, problem-solving with AI and data analytics
Williams et al., 2019	Member	USA	Kindergarten	PopBots	Knowledge-based systems, supervised learning, generative AI
Williams et al., 2021	Member	USA	Middle School	Teachable Machine	AI literacy, machine learning, data collection, and ethical concepts were taught
Williams et al., 2022	Member	USA	Middle and High School	ML Blocks	TinyML, microcontrollers, artificial neural networks
Williams et al., 2024	Member	USA	K-12	Doodlebot System	Perception, representation, and reasoning, AI-based generation
Williams, Alghowinem and Breazeal, 2024	Member	USA	Middle School	AI Playground and Virtual Jibo Robot	Speech-based AI, well-being, ethics

Author and Year	OECD Membership Status	Country Where Study Was Conducted	Educational Level	Artificial Intelligence Tool/Application Used	Taught/Addressed Artificial Intelligence Topics
Williams, Ali et al., 2022	Member	USA	Middle School	Teachable Machine, Creative AI	AI literacy, ethical awareness, critical analysis of AI systems
Yoder et al., 2021	Member	USA	High School	Snap	AI algorithms and enhanced problem-solving skills were taught
Druga et al., 2019	Member	USA, Germany, Denmark, Sweden	Primary School	Cognimates	AI literacy, computational thinking, ethical awareness
Druga and Ko, 2021	Member	USA, Sweden, China	Primary School	Teachable Machine	Machine learning, dataset creation, perception of machine intelligence
Jatzlau et al., 2019	Member	Germany	Middle and High School	Snap!	Reinforcement learning, machine learning, algorithm development
Olari et al., 2021	Member	Germany	K-12	Open Roberta Lab (Neural Network Playground, Q-learning)	Supervised, unsupervised, and reinforcement learning
Seraj et al., 2019	Member	Germany	Primary and Middle School	AI Teaching Assistants	AI perception, teaching aids, computational thinking
Kahn and Winters, 2021	Member	United Kingdom	Middle and High School	Snap!	AI programming, speech and visual recognition, deep learning
Bilstrup et al., 2020	Member	Denmark	High School	Card-Based ML System	ML system design, ethical problems, and solutions
Bilstrup et al., 2024	Member	Denmark	Middle and High School	ml-machine.org, Micro	Teaching ML models, visual programming, AI literacy
Kaspersen et al., 2021	Member	Denmark	High School	VotestratesML, ML Models	Machine learning, data analysis, societal impacts
Schaper et al., 2023	Member	Denmark	Middle School	Teachable Machine	AI literacy, computational empowerment, ethical and societal issues related to technology
Tedre et al., 2021	Member	Finland	Primary and Middle School	Teachable Machine	Machine learning, data analysis, computational thinking
Vartiainen et al., 2020	Member	Finland	Kindergarten and Primary School	Google Teachable Machine	Machine learning, dataset creation
Vartiainen et al., 2021	Member	Finland	Middle School	K-Means Clustering	Machine learning, face recognition, gesture and speech recognition technologies, AI concepts
Park and Shin, 2021	Member	South Korea	K-12	Tooee and Scratch	Big data, machine learning
Park and Shin, 2022 (IEEE)	Member	South Korea	K-12	Scratch and Tooee	Text processing, data collection, and visualization
Y. Park and Shin, 2022	Member	South Korea	K-12	Modified Tooee	Large-scale data training, natural language processing
Cuadrado et al., 2023	Member	Spain	Primary School	Scratch and LSTM (Long Short-Term Memory)	Identification of assistance needs, block-based programming, ethical concerns
Estevez et al., 2019	Member	Spain	High School	Scratch and AI Algorithm Simulations	Machine learning, algorithm design, computational thinking
Garcia et al., 2019	Member	Spain	Primary and Middle School	Scratch, Machine Learning for Kids	Machine learning, classification, data analysis, algorithm design
León et al., 2024	Member	Spain	Primary and Middle School	Scratch, Machine Learning for Kids	Computational thinking, fundamentals of AI
Rodríguez-García et al., 2020	Member	Spain	Middle and High School	LearningML	Supervised learning, data visualization
Rodriguez-Garcia et al., 2021	Member	Spain	Middle School	LearningML	Machine learning, AI literacy
Baldoni et al., 2024	Member	Italy	Primary and Middle School	Teachable Machine	Effects of AI on human and machine minds, cognitive thinking skills

Author and Year	OECD Membership Status	Country Where Study Was Conducted	Educational Level	Artificial Intelligence Tool/Application Used	Taught/Addressed Artificial Intelligence Topics
Kazemitabaar et al., 2023	Member	Canada	K-12	Scratch, Teachable Machine	AI-based STEM projects, data analysis, machine learning
Andersen et al., 2022	Member	Norway	Primary and Middle School	Scratch and CSCL Tools	Collaborative learning, physical computing, human-centered AI
Quiroz and Gutierrez, 2024	Member	Chile	Primary School	Scratch-NB	Naive Bayes classifiers, fundamental machine learning concepts
Sabuncuoğlu, 2020	Member	Türkiye	Middle School	Scratch, Teachable Machine, Python	Algorithm design, data analysis, visual processing, natural language processing
Sabuncuoğlu and Sezgin, 2022	Member	Türkiye	Primary and Middle School	Teachable Machine, Scratch	Visual recognition, object detection, data collection, and model building
Karalekas et al., 2023	Member	Greece	K-12	STEM Robotic Kits	Machine learning, AI processes, and robotics education
Queiroz et al., 2020	Non-Member	Brazil	K-12	BlockWiSARD	AI education, machine learning
Wangenheim et al., 2021	Non-Member	Brazil	K-12	Various Visual Tools	Visual programming, supervised learning
Chen et al., 2024	Non-Member	China	Primary School	ChatScratch	Autonomous programming learning, interactive storytelling, large language models
Deng et al., 2021	Non-Member	China	Middle School	Posture Recognition Algorithms	Pose recognition, computer vision, data processing
Gong et al., 2021	Non-Member	China	K-12	GPU-Supported AI Educational Platforms	AI acceleration, data parallelization, big data analysis
Song et al., 2022	Non-Member	China	K-12	Tencent Coding	Fundamentals of AI, smart campus applications
Song et al., 2023	Non-Member	China	Primary and Middle School	Tencent Coding	AI knowledge structures, ethical and societal impacts
Zhao et al., 2021	Non-Member	China	K-12	AI-Driven Learning Platforms	Personalized learning, computational thinking, data analysis
Cen et al., 2023	Non-Member	China and United Arab Emirates	Middle and High School	Blockly–Electron	Usability, ease of learning, software interface design
Labanda-Jaramillo et al., 2022	Non-Member	Ecuador	Primary and Middle School	Scratch	AI algorithms, machine learning, ethics, and creative thinking skills were taught
Alam, 2022	Non-Member	India	Middle School	Scratch, Alice	Machine learning, classification, game-based algorithms
Kahn et al., 2022	Non-Member	India	K-12	Snap!, eCraft2Learn	Speech recognition, visual processing, natural language processing, deep learning, AI ethical issues
Priya et al., 2022	Non-Member	India	Middle and High School	ML-Quest	Supervised learning, K-Nearest Neighbors, gradient descent
Vazhayil et al., 2019	Non-Member	India	Middle and High School	ML4Kids	AI literacy, data literacy, computational thinking skills were taught
Chiu et al., 2021	Non-Member	Hong Kong	Middle School	AI4Future	AI perception, machine learning, and ethics were taught
Chiu, 2021	Non-Member	Hong Kong	K-12	Scratch	AI knowledge and processes, societal impacts of AI, teacher-student communication
Dai, 2024	Non-Member	Hong Kong	Primary and Middle School	Machine Learning	AI literacy, machine learning, large language models, ethical awareness
Ng et al., 2023	Non-Member	Hong Kong	Middle School	Maker Education Tools	AI literacy, problem-solving, collaborative learning skills with recycling projects
Su and Yang, 2023	Non-Member	Hong Kong	Kindergarten	AI4KG	Perception of robots, perceptions of engineering and science

Author and Year	OECD Membership Status	Country Where Study Was Conducted	Educational Level	Artificial Intelligence Tool/Application Used	Taught/Addressed Artificial Intelligence Topics
Su and Zhong, 2022	Non-Member	Hong Kong	Kindergarten	PopBots	Basic AI concepts, problem-based learning
Shamir and Levin, 2022	Non-Member	Israel	Primary School	PopBots	Supervised learning, classification, data analysis
Sanusi et al., 2023	Non-Member	Nigeria	Middle School	AI MyData	Ethical aspects of machine learning, societal impacts
Alturayeif et al., 2020	Non-Member	Saudi Arabia	Primary and Middle School	DeepScratch	Deep learning algorithms, AI was taught
Aung et al., 2022	Non-Member	Thailand	Middle School	AIThaiGen	Building and testing AI models, ethical awareness
Huang, 2024	Non-Member	Taiwan	K-12	Scratch, Generative AI Tools	Visual programming, generative AI, programming logic
Weng et al., 2024	Non-Member	Taiwan	Primary School	Scratch, AI-Supported Comic Platform	Sustainable development, empathy development, animation
Hsu et al., 2021	Non-Member	Taiwan and USA	Middle School	MIT App Inventor, Personal Image Classifier	Image recognition, programming logic, AI-STEAM integration
Hsu et al., 2022	Non-Member	Taiwan and USA	Middle School	MIT App Inventor, Speech-Based AI Tools	Speech-based AI, voice user interface, computational thinking
Valko et al., 2022	Non-Member	Ukraine	Middle and High School	Google Teachable Machine, DALL-E, PictoBlox	Machine learning, cloud-based data analysis techniques

Table 1 summarizes the main characteristics of the studies included in this review. It provides information on the country where the study was conducted, the educational level, and the AI tools used. This overview helps readers quickly understand which educational levels utilize specific AI applications.

Table 2. Descriptive information about the studies

Author and Year	Recommendations (Target Audience Emphasis)	Pedagogical Impacts
Akram et al., 2022	It is recommended to develop supplementary materials and enhance teacher training to improve the development of future lesson plans.	Significant improvements were observed in understanding AI concepts; however, additional support materials were needed for some students.
Basu, 2019	It is advised to determine the areas where students require support and provide more training to teachers on how to effectively use rubrics.	Rubrics were effective in identifying differences in student skills; it was determined that some students needed support in specific coding topics.
Broll and Grover, 2023	Implementing phased learning activities at various educational stages and researching their long-term impacts is recommended.	Students were able to better comprehend machine learning (ML) algorithms and gained deeper knowledge in coding processes; awareness of ethics and bias increased.
Chung and Shamir, 2021	Longer-term monitoring and repeating machine learning teaching using different robotic systems are suggested.	A significant improvement was recorded in students' knowledge and understanding of machine learning topics, and the contribution of robots to learning was observed.
Grover et al., 2023	Expanding applications with more extensive teacher and student involvement is recommended.	It was observed that students gained a deeper understanding of cybersecurity and AI topics; additionally, awareness of ethics and bias increased.
Grover, 2024	Conducting additional empirical research and providing support to teachers in artificial intelligence education are advised.	Comprehensive recommendations were provided on the current state and future trends of K-12 AI education; guidance outputs were offered for AI education.
Irgens et al., 2022	Applying a critical machine learning perspective in various learning settings and examining its long-term effects are recommended.	Children established more sophisticated connections regarding social and ethical orientations and gained skills related to machine learning.
Jayasuriya et al., 2024	It is suggested to extend STEAM-focused AI education to a larger number of educators.	Teachers achieved higher self-confidence and teaching capacities in AI and computer vision technologies.
Jordan et al., 2021	Investigating the prolonged impacts of student projects and applying them in diverse learning contexts are advised.	Students better understood body movements and interactive AI systems; awareness of ethical issues increased.

Author and Year	Recommendations (Target Audience Emphasis)	Pedagogical Impacts
Kim and Kwon, 2024	Expanding the use of tangible computing tools to more extensive student groups and creating training programs for educators are recommended.	Students educated with tangible computing tools had increased understanding and perception levels of AI; tangible computing tools were effective in this process.
Kumar and Worsley, 2023	It is suggested to evaluate sports-oriented AI projects with more substantial student populations and across various sports fields.	Students gained skills in developing AI-based applications and working with sports data; programming and data analysis abilities increased.
Kurz et al., 2024	Reapplying practices in varied educational settings with increased teacher and student numbers is recommended.	The use of Teachable Machine increased teachers' technology integration and positively affected students' interest in science classes.
Lee et al., 2021	Researching the impacts of AI-assisted game-based learning approaches across different ages and educational stages is advised.	The game-based learning method contributed to students gaining a deeper understanding of life sciences and AI topics and developing problem-solving skills.
Long et al., 2022	Enhancing support for familial learning experiences and broadening such programs are suggested.	Family collaboration in AI literacy activities was effective in children's learning processes, and communication skills developed.
Okolo, 2024	It is recommended to extend AI education to broader student audiences and enhance its integration into curricula.	Students participating in AI workshops increased their knowledge and awareness in AI topics; notably, their ethical awareness improved.
Park et al., 2021	It is advised that visual interfaces be more extensively utilized in primary education.	The user interface supported learning; however, some misunderstandings occurred.
Park et al., 2022	The broadening of game-based educational settings is encouraged.	Game-based learning increased student interest.
Percival et al., 2022	Future research should consider involving more significant numbers of students for testing purposes.	Cryptography concepts were understood with a 60% success rate; overall satisfaction was at 90%.
Reddy et al., 2021	Proposals include enhancing practical exercises and advancing model creation.	Students showed improvement in text classification and natural language processing topics.
Reddy et al., 2022	It is recommended to adopt automated evaluation instruments more broadly.	Improvements in project quality were observed based on feedback.
Taylor et al., 2019	Enhancing instructional resources and increasing teacher professional development are advised.	Student interest increased when block-based programming was combined with game-based environments.
Touretzky et al., 2019	Incorporating the five core concepts of AI into educational programs is recommended.	It was determined that AI education should become more widespread at the K-12 level.
Touretzky et al., 2023	Strengthening curriculum alignment and providing additional training for teachers are suggested.	An advanced conceptual framework for AI education was proposed.
von Wangenheim et al., 2021	Future research should include extensive ethical debates and practical engagements.	Students gained knowledge about speech-based AI systems but struggled with ethical issues.
Walsh et al., 2023	Implementing hands-on activities to more effectively convey ethical issues is advised.	Students developed deeper ethical understandings of AI, and teachers employed strategies to adapt the curriculum.
Wan et al., 2020	Widening the inclusion of AI-assisted scientific exploration projects within the curriculum is recommended.	AI tools enhanced students' computational thinking skills during scientific discovery processes, and they created innovative projects.
Williams et al., 2019	Promoting artificial intelligence education among younger learners is suggested.	Children's perceptions of AI changed depending on age, and they were positively influenced by the activities.
Williams et al., 2021	Increasing the incorporation of AI initiatives within classroom settings is advised.	AI projects improved students' understanding of AI and their ethical awareness.
Williams et al., 2022	Encouraging the use of tangible instruments in AI learning is recommended.	Students gained greater understanding in developing TinyML models during the process.
Williams et al., 2024	Implementing extended interventions covering a range of subjects is suggested.	Students were successful in learning AI topics and developing creative projects.
Williams, Alghowinem and Breazeal, 2024	Increasing experiential activities that emphasize social well-being is recommended.	Students learned more about AI-based chatbots and were able to discuss ethical issues.
Williams et al., 2022	It is advised to broaden AI education by emphasizing ethical considerations.	Ethical awareness development and improvements in critical perspectives on AI systems were observed in students who received project-based AI education.

Author and Year	Recommendations (Target Audience Emphasis)	Pedagogical Impacts
Yoder et al., 2021	Highlighting the importance of applying AI algorithms to practical, real-life challenges is recommended.	Students improved their abilities to learn the Breadth-First Search (BFS) algorithm, and their problem-solving skills increased through the use of AI tools.
Druga et al., 2019	Enhancing the accessibility of inclusive educational activities for economically disadvantaged groups is suggested.	Children from low and middle socio-economic groups developed more collaboration skills despite having less experience, but they struggled to progress.
Druga and Ko, 2021	Developing innovative educational approaches tailored to help children comprehend AI more effectively is advised.	After AI education, children better understood machine intelligence and became more conscious in coding processes.
Jatzlau et al., 2019	Utilizing reinforcement learning tools across various educational stages and studying their long-term learning outcomes is recommended.	Significant increases were observed in students' knowledge and understanding levels of machine learning and reinforcement learning concepts.
Olari et al., 2021	Augmenting the number of robotics-related activities within AI education is suggested.	Students better understood the AI learning paradigm.
Seraj et al., 2019	Expanding the deployment of instructional aides and AI technologies in educational environments is advised.	AI-supported teaching aids increased students' interest in lessons and improved their computational thinking skills.
Kahn and Winters, 2021	Assessing the sustained impacts of diverse projects with more extensive student cohorts is recommended.	Significant improvements were seen in students' knowledge and skill levels regarding AI concepts; however, some students had difficulty developing projects.
Bilstrup et al., 2020	It is suggested to involve more students and examine workshop implementations across varied educational settings.	Notable developments were observed in ethical thinking processes; students could better analyze ethical issues in ML system design.
Bilstrup et al., 2024	Future studies should test applications with broader participant demographics and across different educational frameworks.	Teachers were observed to use the platform effectively, and students were able to better comprehend ML concepts.
Kaspersen et al., 2021	Researching the social implications of machine learning technologies and their use in diverse learning contexts is advised.	Significant improvements were seen in students' understanding of machine learning and its societal impacts; however, some students struggled with complex concepts.
Schaper et al., 2023	Offering design recommendations and tangible examples for upcoming initiatives is recommended.	Youths' understanding of AI/ML technologies developed through concepts of human-like intelligence; technical and societal awareness increased.
Tedre et al., 2021	Increasing practical engagements and enhancing teacher education are suggested.	Students better understood machine learning topics.
Vartiainen et al., 2020	Conducting extensive research and applications involving a larger number of participants is recommended.	Children made progress in understanding machine learning concepts through quick and concrete interactions.
Vartiainen et al., 2021	Broadening machine learning initiatives to include additional students is advised.	Students improved their understanding of ML and application ideas, making progress in grasping basic principles related to ML.
Park and Shin, 2021	Implementing further practical exercises and extended interventions is recommended.	An increase was observed in students' data collection and analysis skills.
Park and Shin, 2022 (IEEE)	Engaging in more profound exploration of natural language processing and text mining is advised.	Significant improvements were observed among students in text processing topics.
Y. Park & Shin, 2022	Addressing the complexities associated with handling extensive data collections is recommended.	The system received high scores in terms of usability and scalability.
Cuadrado et al., 2023	Continuously evaluating the model's efficacy across various educational settings and with more substantial datasets is advised.	The gender factor had a significant effect on the model's performance; accuracy in identifying students needing help increased.
Estevez et al., 2019	Implementing AI instruction using block-based platforms such as Scratch across multiple educational stages is recommended.	Workshop activities improved students' understanding of AI and enhanced their computational thinking skills.
Garcia et al., 2019	Future research should consider executing diverse AI projects with more significant student populations.	A significant improvement was recorded in students' levels of understanding machine learning concepts; problem-solving skills increased.

Author and Year	Recommendations (Target Audience Emphasis)	Pedagogical Impacts
León et al., 2024	Offering strategic guidance for embedding AI within the K-12 educational framework is advised.	Computational thinking and AI literacy improved; a gender gap was observed.
Rodríguez-García et al., 2020	Proposals include expanding implementations and enhancing teacher support mechanisms.	Students showed progress in computational thinking and developing AI projects.
Rodriguez-Garcia et al., 2021	Promoting the inclusion of LearningML in educational programs is recommended.	Improvement was observed in students' basic understanding of AI.
Baldoni et al., 2024	Future research should examine the outcomes of extended monitoring and varied instructional methodologies.	Significant improvement was seen in children's perceptions of AI, but some misconceptions persisted.
Kazemitabaar et al., 2023	Studying the feasibility of experiential learning techniques across diverse ages and educational stages is advised.	Meaningful learning opportunities were provided to students through AI projects in STEM classes; conceptual knowledge levels increased.
Andersen et al., 2022	Reiterating practices with more substantial student groups and analyzing prolonged impacts in future research is recommended.	An increase was observed in students' collaborative learning skills; they developed a better understanding of Human-Centered AI (HCAI).
Quiroz & Gutierrez, 2024	Modifying Scratch-NB to address more sophisticated subjects is suggested.	Eighty percent of participants were successful in learning to use Scratch-NB.
Sabuncuoğlu, 2020	Enhancing educational programs through additional teacher development and varied instructional resources is advised.	Significant improvement was detected in students' problem-solving and algorithmic thinking skills; student interest and participation were high.
Sabuncuoğlu and Sezgin, 2022	Evaluating the effectiveness of AI tools with different student demographics over extended durations is recommended.	Children more concretely grasped AI concepts and could make connections between programming and the physical world.
Karalekas et al., 2023	It is suggested to integrate AI and robotics more comprehensively into curricula and increase investments in educator training.	AI comprehension and algorithm processes were effectively taught using robotic tools; students improved in STEM-focused skills.
Queiroz et al., 2020	Increasing hands-on experiences within AI education is recommended.	Students were enabled to understand the differences between AI and traditional programming.
Wangenheim et al., 2021	Enhancing tool functionalities to boost student participation is advised.	Visual tools increased ML understanding, but additional educational design was needed.
Chen et al., 2024	Future research should replicate studies with more extensive cohorts and across various age brackets.	Higher creativity and skill development were recorded in students' programming processes; the interactive support system enhanced the learning experience.
Deng et al., 2021	It is recommended to execute diverse AI initiatives with broader student involvement in forthcoming research.	Significant increases were observed in students' knowledge levels on computer vision and AI topics; project-based learning contributed to cognitive skills.
Gong et al., 2021	Examining the efficiency of GPU-accelerated platforms across various technologies and educational contexts is advised.	GPU-supported educational platforms increased learning speed and efficiency; students better understood big data concepts.
Song et al., 2022	Enhancing teacher education and developing curricular content are recommended.	AI education increased student interest and developed computational thinking skills.
Song et al., 2023	Creating additional materials and support for educators in AI instruction is advised.	Students better understood and applied AI concepts.
Zhao et al., 2021	Incorporating AI-driven educational platforms into programs serving extensive student populations is recommended.	AI positively affected student performance in personalized learning processes, providing more efficient learning outcomes.
Cen et al., 2023	Future studies should perform usability evaluations of various software interfaces with larger groups of students.	Significant improvements were observed in learning ease and software utility; the user-friendly interface increased satisfaction.
Labanda-Jaramillo et al., 2022	Widening the incorporation of AI studies into additional educational levels is suggested.	Positive developments were observed in children's understanding of AI concepts, contributing to the learning process.
Alam, 2022	Assessing game-based learning methods across various age groups and studying their prolonged impacts is recommended.	A significant increase was observed in students' knowledge levels regarding AI concepts; the positive effects of digital games on learning motivation were recorded.
Kahn et al., 2022	Researching how Snap! blocks can be used in various AI subjects and their lasting educational effects is advised.	Students gained a deeper understanding of AI and related technologies, but some difficulties were experienced regarding ethical and privacy issues.

Author and Year	Recommendations (Target Audience Emphasis)	Pedagogical Impacts
Priya et al., 2022	Offering guidance to help students progress to higher-level machine learning concepts is recommended.	Game-based learning was observed to enhance learning performance.
Vazhayil et al., 2019	Enhancing teacher education through additional digital tools is advised.	Challenges in the adoption of AI education among teachers were noted; however, positive effects of game-based approaches were reported.
Chiu et al., 2021	Investigating the motivational aspects and sustained outcomes of AI learning is recommended.	Positive developments were observed in students' perceptions of AI, and an increase in learning motivation was detected.
Chiu, 2021	Applying various AI instructional models within an extensive educational system and contrasting their results is suggested.	The holistic approach offered a more sustainable model by considering different dimensions of AI education; student and teacher participation increased.
Dai, 2024	Utilizing the dual-contrast teaching method across multiple ages and educational tiers is advised.	The experimental group showed significant increases in AI knowledge and skills; learning motivation increased, and learning anxiety decreased.
Ng et al., 2023	Broadening AI and maker-focused learning through practical applications is recommended.	Project-based learning activities increased students' AI literacy and interest in recycling projects.
Su and Yang, 2023	Undertaking additional studies and assessments to comprehend enduring impacts is suggested.	Children's perceptions of robots improved, but no significant changes were observed in attitudes toward engineering and science.
Su and Zhong, 2022	Encouraging the use of problem-solving tasks and interactive experiences is advised.	Children made progress in understanding AI concepts.
Shamir and Levin, 2022	Suggesting increased practice opportunities and mentorship for advanced education is recommended.	Students advanced in computational thinking and data analysis skills.
Sanusi et al., 2023	Promoting the growth of AI instruction within African educational environments is advised.	Students made progress in machine learning and ethical awareness.
Alturayeif et al., 2020	Enhancing the incorporation of deep learning concepts within Scratch is recommended.	Students better understood AI concepts using Scratch; creative advancements were made in projects.
Aung et al., 2022	Advocating for the widespread use of AIThaiGen in AI learning and extending its application beyond school settings is suggested.	Significant improvement was observed in students' AI understanding who were educated with AIThaiGen; learning motivation increased.
Huang, 2024	Evaluating generative AI and visual programming tools across various learning contexts and age demographics is recommended.	Visual programming tools and generative AI were effective in developing students' programming skills; however, some students experienced difficulties.
Weng et al., 2024	Enhancing the interactive nature of education and boosting student engagement is advised.	Students made progress in scriptwriting and animation skills, but no significant difference was observed in empathy levels.
Hsu et al., 2021	Studying the feasibility of hands-on learning strategies across different ages and education levels is recommended.	Significant progress was observed in students' image recognition and programming skills; experiential learning was effective.
Hsu et al., 2022	Implementing various experiential learning methods with more extensive participant groups is advised.	Female students performed better with experiential learning methods; speech-based AI contributed to learning.
Valko et al., 2022	Creating additional cloud-based instruments and materials for AI instruction is recommended.	AI education supported by cloud technologies increased student engagement and learning performance.

Table 2 highlights the pedagogical effects and recommendations reported in the studies. It shows how students' learning outcomes and teaching processes were impacted. Additionally, the table offers guidance for improving AI integration in education, based on the lessons drawn from previous research.

AI Tools Used According to the Educational Level Variable

The employment of visual and block-based programming tools at various educational levels has demonstrated diversity that is suitable for the age and cognitive development stages of the students. In particular, the resources utilized, and the subjects of artificial intelligence taught at the elementary, middle, and high school levels vary depending on the educational level. The AI tools and subjects taught vary by educational stage, as summarized in Table 3.

Table 3. Tools and Platforms Used in AI Education According to Educational Level Factors

Education Level	Frequency(f)	Primary Artificial Intelligence Tools (Usage Count)
Preschool	3	PopBots (2), AI4KG (1)
Elementary School	11	Scratch (5), PRIMARYAI (3), AI-Supported Comic Platform (1), ChatScratch (1), Tangible Computing Tools (1), Teachable Machine (2), Tooee (2)
Elementary & Middle School	14	Scratch (4), Teachable Machine (3), AI-Supported Comic Platform (1), AI Teaching Assistants (1), BlockWiSARD (1), Cognimates (1), DeepScratch (1), K-Means Clustering (1), Machine Learning for Kids (2), Open Roberta Lab (1), PRIMARYAI (2)
Middle School	32	Scratch (14), Teachable Machine (8), Snap! (3), NetsBlox (2), AI4Future (1), AI MyData (1), AI Playground (1), AIThaiGen (1), Alice (1), BlockWiSARD (1), Blockly–Electron (1), Cognimates (1), Creative AI (1), CryptoScratch (1), DeepScratch (1), Generative AI Tools (1), IMU-based Motion Detection Tools (1), IntelliBlox (1), MIT App Inventor (1), ML-Quest (1), Open Roberta Lab (1), Personal Image Classifier (1), Pose Detection (1), Posture Recognition Algorithms (1), Speech-Based AI Tools (1), TensorFlow (1), Unity (1), Virtual Jibo Robot (1)
Middle & High School	7	Snap! (1), AI Ethics Modules (1), Blockly–Electron (1), DALL-E (1), Generative AI Tools (1), Google Teachable Machine (1), LearningML (1), ML Blocks (1), ML-Quest (1), PictoBlox (1)
High School	7	NetsBlox (2), Snap! (2), AI Ethics Modules (1), Blockly–Electron (1), Card-Based ML System (1), DALL-E (1), Google Teachable Machine (1), LearningML (1), ML-Quest (1), PictoBlox (1), VotestratesML (1)
K-12	21	Scratch (3), AI4K12 (2), Teachable Machine (2), Tencent Coding (2), AI-Driven Learning Platforms (1), AI4Future (1), BlockWiSARD (1), eCraft2Learn (1), Generative AI Tools (1), Google's Quick, Draw! (1), GPU-Supported AI Educational Platforms (1), LevelUp (1), ML4Kids (1), Modified Tooee (1), PopBots (1), PRIMARYAI (1), SmileyCluster (1), Snap! (1), STEM Robotic Kits (1), TensorFlow Playground (1), Tooee (1), Various Visual Tools (1)

Note. The total frequency (95) exceeds the number of unique studies (91) because some studies target multiple educational levels (e.g., K-12, Elementary & Middle School).

Table 3 presents the distribution of AI tools across different educational levels. It indicates the frequency of use and which tools were preferred at each level. This helps illustrate how AI applications vary according to students' cognitive development and learning needs.

Scratch, PopBots, Teachable Machine, and Machine Learning for Kids are the most often utilized tools in elementary school-level research (Table 3). Students can improve their algorithmic thinking skills and learn the fundamentals of programming with Scratch (García et al., 2019; Moreno-León et al., 2024). PopBots give kids a tangible and participatory approach to learn basic AI concepts including data analysis, classification, and supervised learning (Su & Zhong, 2022; Williams et al., 2019). According to Teachable Machine studies, students acquire expertise in dataset construction and machine learning (Vartiainen et al., 2020). Furthermore, students' views and comprehension of AI have improved because of the employment of tangible computing tools (Kim & Kwon, 2024).

Tools like Scratch, Teachable Machine, Snap!, Blockly, AI4Future, CryptoScratch, and Machine Learning for Kids are widely utilized at the middle school level (Table 3). Natural language processing, data analysis, algorithms, machine learning, ethical awareness, and computational thinking are all skills that these technologies help students develop (Alam, 2022; Irgens, Vega, Adisa, & Bailey, 2022; Baldoni et al., 2024). For example, symmetric and asymmetric encryption techniques were taught using CryptoScratch, and students' comprehension of cryptography principles increased (Percival, Rayavaram, Narain, & Lee, 2022). Furthermore, it was discovered that the AI4Future curriculum improved students' understanding of AI, machine learning, and ethical concerns (Chiu et al., 2021).

More sophisticated technologies like NetsBlox, Snap!, IntelliBlox, Unity, and VotestratesML are utilized in high school academics (Table 3). These resources allow students to study advanced subjects like big data education, ethical awareness, data analysis, machine learning algorithms, and AI-based cybersecurity (Broll & Grover, 2023; Kaspersen et al., 2021; Yoder et al., 2021). For instance, it was noted that students' awareness of ethics and bias rose along with their comprehension of cybersecurity and AI subjects when they used NetsBlox (Grover, Broll, & Babb, 2023).

Tools like Scratch, Tooee, MIT App Inventor, Teachable Machine, and AI4K12 are employed in research spanning the K–12 spectrum (Table 3). By providing content appropriate for students of various ages, these resources aid in the teaching of AI

principles over a broad spectrum (Grover, 2024; Touretzky et al., 2019). By outlining the five major concepts of artificial intelligence, the AI4K12 effort seeks to include these ideas into the K–12 curriculum (Touretzky et al., 2019).

Recommendations to Researchers from the Conducted Studies

Several suggestions for future researchers have been made in the results and discussion parts of the examined publications, highlighting study gaps.

Research conducted at the elementary school level highlights the need for increased emphasis on early AI education, the general use of visual interfaces, and the encouragement of family participation (Long et al., 2022; Su & Yang, 2023). For example, Long et al. (2022) suggest developing family-based learning programs, observing that such programs enhance children's communication and learning skills. In addition, Kim and Kwon (2024) recommend implementing teacher training programs and utilizing tangible computing tools with larger student groups to strengthen students' engagement and understanding of AI concepts. These findings are consistent with Table 2, which illustrates the pedagogical impact of block-based AI tools on students' cognitive and affective development at the elementary level.

Middle school studies advocate for increased practical applications, enhanced teacher support, and the adoption of game-based and project-based learning approaches, alongside a deeper consideration of ethical issues (Alam, 2022; Irgens et al., 2022; Baldoni et al., 2024). Alam (2022) found that participation in digital game-based learning increased students' motivation and improved their grasp of AI principles. Similarly, Irgens et al. (2022) emphasize that integrating hands-on AI projects can foster both computational and ethical understanding, reinforcing the outcomes reported in Table 2.

At the high school level, research suggests prioritizing cybersecurity, ethical considerations, and real-world applications of AI education. Expanding workshops, project-based initiatives, and examining the long-term effects of AI instruction are also recommended (Broll & Grover, 2023; Kaspersen et al., 2021). For instance, Broll and Grover (2023) argue that staged learning exercises should be employed at different educational levels, and their long-term outcomes systematically evaluated.

Studies encompassing the K–12 spectrum highlight the need to integrate AI education into STEM, robotics, and big data-based projects while increasing teacher preparation and curriculum inclusion for students of diverse ages (Grover, 2024; Touretzky et al., 2019). The AI4K12 framework proposed by Grover (2024) provides a roadmap for implementing AI across grade levels, whereas Touretzky et al. (2019) stress the incorporation of five core AI concepts into curricula to ensure continuity and conceptual depth.

Across all levels, the need for additional teacher training, instructional resources, and interactive, hands-on learning environments is emphasized (Irgens et al., 2022; Alam, 2022). Broll and Grover (2023) further recommend integrating ethical considerations and discussions of bias into AI curricula. Finally, accessibility remains a challenge, particularly in non-OECD countries, where resource constraints and limited technological infrastructure may hinder equitable AI education. Druga et al. (2019) highlight the importance of designing inclusive learning activities to reach students from low socio-economic backgrounds.

Pedagogical Impacts of the Conducted Research

Block-based AI instruction has a range of pedagogical effects on students' cognitive, affective, and psychomotor development, with noticeable differences across educational levels.

Research in primary schools shows that students achieved significant gains in knowledge and skills in areas such as animation, machine learning, data analysis, algorithm design, and empathy development (García et al., 2019; Moreno-León et al., 2024). For example, a study using Scratch and Machine Learning for Kids reported substantial improvements in students' computational thinking and AI literacy (Moreno-León et al., 2024). Furthermore, Kim and Kwon (2024) found that learners who received instruction with tangible computing tools demonstrated higher comprehension and deeper understanding of AI concepts. These findings align with the data summarized in Table 2, highlighting the effectiveness of interactive, block-based tools in fostering foundational AI skills at the elementary level.

At the middle school level, students' knowledge and abilities in machine learning, algorithms, block-based programming, data analysis, natural language processing, ethical reasoning, and computational thinking showed marked improvements (Alam, 2022). As indicated in Table 2, pupils acquired machine learning competencies and developed complex connections between social and ethical considerations (Irgens et al., 2022). However, some students faced challenges in project development and conceptual understanding (Baldoni et al., 2024), emphasizing the need for targeted support and differentiated instruction.

High school research further demonstrates that students enhanced their computational thinking, deepened their understanding of AI and machine learning concepts, and became more ethically aware (Broll & Grover, 2023; Kaspersen et al., 2021). For instance, Yoder et al. (2021) reported that students' algorithmic thinking and problem-solving skills improved when using Snap! to teach the BFS algorithm through real-world scenarios. Moreover, learners developed a more nuanced understanding of ethical implications and societal impacts of AI.

Studies spanning the K–12 spectrum underscore the significant role of AI education in advancing students' technical and cognitive skills (Grover, 2024; Touretzky et al., 2019). Block-based and visual tools simplify the learning of complex AI concepts, increasing engagement and motivation. Nonetheless, challenges such as correcting misconceptions, addressing ethical and societal issues, and providing sufficient teacher training remain (Von Wangenheim, Hauck, Pacheco, & Bertonceli Bueno, 2021).

Distribution of Conducted Research by Year and Country

Block-based AI education has garnered interest worldwide and is especially prevalent in OECD member nations, according to an analysis of the distribution of completed research by years and nations.

Table 4. Countries Where Studies Were Conducted According to Year and OECD Membership Status

Year	OECD Membership Status	Country (Frequency(f))	Year
2019	Member	USA (4), Germany (2), USA-Germany-Denmark-Sweden (1), Spain (2)	2019
2020	Member	USA (1), Denmark (1), Finland (1), Spain (2), Türkiye (1)	2020
2021	Member	USA (7), USA-Sweden-China (1), Germany (1), United Kingdom (1), Denmark (1), Finland (2), South Korea (1), Spain (1)	2021
2022	Member	USA (8), South Korea (2), Norway (1), Türkiye (1)	2022
2023	Member	USA (5), Denmark (1), Spain (1), Canada (1), Greece (1)	2023
2024	Member	USA (7), Denmark (1), Spain (1), Italy (1), Chile (1)	2024
2019	Non-Member	India (1)	2019
2020	Non-Member	Brazil (1), Saudi Arabia (1)	2020
2021	Non-Member	Brazil (1), China (3), Hong Kong (1), Taiwan-USA (1)	2021
2022	Non-Member	China (1), Ecuador (1), India (3), Hong Kong (1),Ukraine(1), Israel (1), Thailand (1), Taiwan and USA (1)	2022
2023	Non-Member	China (1), China and United Arab Emirates (1), Hong Kong (2), Nigeria (1)	2023
2024	Non-Member	China (1), Hong Kong (1), Taiwan (2)	2024

As shown in Table 4, the number of studies conducted has steadily increased since 2019. Both OECD member and non-member nations experienced a notable rise in publications, particularly in 2021 and 2022, indicating growing scholarly interest in AI education as its global importance is increasingly recognized.

Compared to non-OECD nations, there are more, and better studies carried out in OECD member nations. It has been shown that nations with more developed curriculum integration of AI education, teacher training, and technology infrastructure include the United States, Germany, Denmark, Finland, South Korea, Spain, and Türkiye (Grover, 2024; Touretzky et al., 2019). For instance, research in the United States has shown that professional development programs have improved instructors' self-confidence and their ability to teach AI and computer vision technologies (Jayasuriya et al., 2024).

It has been observed that the scope and efficacy of AI education are impacted in non-OECD nations by elements including a lack of funding and a lackluster technological infrastructure (Alam, 2022; Sanusi et al., 2023). Research from China, India, Hong Kong, Nigeria, and Taiwan has shown that although AI education has helped students advance their technical and cognitive abilities, teachers still require more assistance and training (Chiu et al., 2021; Su & Yang, 2023).

CONCLUSION AND RECOMMENDATIONS

The purpose of this study was to methodically examine the pedagogical effects of visual and block-based programming tools in K–12 artificial intelligence (AI) education. Drawing on 91 studies published between 2019 and 2024 across a range of countries, educational levels, and cultural contexts, the findings highlight the significant role of these tools in supporting students' comprehension of AI concepts. Visual and block-based programming platforms were found to vary according to educational levels. For instance, Teachable Machine, PopBots, and Scratch are predominantly used in elementary schools, where they facilitate foundational learning in programming and algorithmic thinking (García et al., 2019; Kim & Kwon, 2024). At the middle school level, more advanced platforms such as Snap!, Blockly, and AI4Future support students in understanding complex topics including data analysis, machine learning, and ethical reasoning (Irgens et al., 2022; Chiu et al., 2021). High school students, on the other hand, benefit from tools like NetsBlox and IntelliBlox, which allow a deeper exploration of sophisticated AI concepts (Broll & Grover, 2023).

The pedagogical effects of these tools demonstrate clear educational benefits. Elementary students show notable improvements in computational thinking, problem-solving, and creativity (Moreno-León et al., 2024). At the middle school level, students' skills in machine learning, data analysis, and ethical awareness improve significantly (Irgens et al., 2022), while high school students gain advanced comprehension of complex AI topics alongside heightened ethical reasoning (Broll & Grover, 2023). These outcomes collectively suggest that AI instruction using visual and block-based programming tools supports cognitive, affective, and psychomotor development, reinforcing the value of these tools across different educational stages.

The study also emphasizes broader structural and contextual factors influencing AI education. Most research focuses on OECD countries, where technological infrastructure, teacher preparation, and curriculum integration are more advanced (Grover, 2024; Touretzky et al., 2019). In contrast, non-OECD nations face challenges such as limited funding and lower technological readiness, constraining the implementation and effectiveness of AI instruction (Druga et al., 2019). Despite these challenges, there is growing interest and emerging applications in these regions (Chiu et al., 2021; Su & Yang, 2023), highlighting the global relevance and potential expansion of AI education.

Across educational levels, the results are consistent with prior research demonstrating that visual and block-based programming tools enhance students' computational thinking, problem-solving, and ethical reasoning (García et al., 2019; Moreno-León et al., 2024; Irgens et al., 2022). Contrastingly, some studies suggest these tools are less effective for conveying highly advanced AI concepts, particularly at the high school level (Baldoni et al., 2024). By situating these findings alongside similar and differing results in the literature, it becomes evident that AI education's effectiveness is shaped not only by the tools themselves but also by teacher preparedness, curriculum integration, and contextual factors such as technological infrastructure, emphasizing the need for holistic implementation strategies.

The analysis also draws attention to pedagogical recommendations for improving AI education. Strengthening teacher training and providing well-structured instructional materials remain critical (Irgens et al., 2022; Grover, 2024). Moreover, integrating ethical considerations of AI into the curriculum and developing immersive, hands-on learning experiences can enhance student engagement and understanding (Alam, 2022; Broll & Grover, 2023). Visual and block-based programming tools show considerable promise in fostering problem-solving and critical thinking (Williams et al., 2019; García et al., 2019), although certain studies indicate that these tools may be less effective for conveying highly complex AI concepts (Baldoni et al., 2024).

In conclusion, the successful implementation of AI education at the K–12 level depends on a strategic combination of visual and block-based programming tools, teacher support, and thoughtfully designed learning environments. These tools not only help students grasp AI principles but also contribute to the development of cognitive, emotional, and psychomotor competencies. Future research should continue exploring the long-term impacts of AI education, focusing on diverse contexts and student populations, while integrating ethical and societal dimensions into pedagogy to ensure comprehensive AI literacy for all learners.

LIMITATIONS

This study provides a comprehensive overview of research on visual and block-based programming tools in K–12 AI education; however, there are several limitations to consider. Firstly, the review only includes studies published between 2019 and 2024.

Secondly, the analysis included journal articles and conference proceedings but excluded unpublished works such as theses, dissertations, and technical reports. This limitation may result in missing findings from practical implementations, local projects, or emerging tools that have not yet reached formal publication.

Thirdly, the studies were unevenly distributed geographically, with a large proportion focusing on OECD member countries. These nations generally have better technological infrastructure, teacher training, and curriculum integration, which may not reflect conditions in non-OECD countries. Consequently, the generalizability of the findings to countries with limited resources, funding, or technological access is constrained.

Fourthly, many studies measured short-term learning outcomes and improvements in computational thinking, problem-solving, or AI understanding. Long-term effects on student motivation, skill retention, and ethical awareness were not consistently assessed, limiting our understanding of the sustained impact of block-based AI education.

Finally, the analyzed research often focused on specific student populations, typically from urban or middle-class backgrounds. Less is known about the effectiveness of these tools for diverse socioeconomic, cultural, or special education contexts, which suggests that the findings may not be fully applicable to all K–12 learners.

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Statements of Publication Ethics

The authors affirm that this study involves no unethical practices and that all stages of the research were conducted in full compliance with established research and publication ethics.

Researchers' Contribution Rate

Both authors contributed equally to this study. Hüseyin Sıhat led the systematic review process, including literature search, data extraction, coding procedures, and data analysis. Mehmet Akif Ocak supervised the overall research process, contributed to

the methodological design, and supported the interpretation of the findings. Both authors jointly drafted, revised, and finalized the manuscript and approved the final version of the article.

Ethics Committee Approval Information

This study is a systematic review and does not involve human participants, interventions, experiments, or the collection of personal or sensitive data. No primary data were collected. In line with national and international ethical standards, systematic reviews do not require ethics committee approval; therefore, ethics approval was not sought.

Use of Artificial Intelligence

During the preparation of this manuscript, artificial intelligence tools ChatGPT-4o and Grammarly were utilized exclusively for the purposes of language editing, proofreading, and enhancing the clarity of the text. These tools assisted in refining sentence structures and ensuring grammatical accuracy. They were not involved in the systematic search strategy, data extraction, data synthesis, or the formulation of the study's conclusions. The data obtained from the AI tools has been rigorously reviewed in accordance with accepted ethical and academic standards. The AI tools are not credited as authors; the authors are solely responsible for the content of the published article.

REFERENCES

Alam, A. (2022). A digital game based learning approach for effective curriculum transaction for teaching-learning of artificial intelligence and machine learning. In *2022 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS)* (pp. 473–478). IEEE. https://doi.org/10.1109/ICSCDS54817.2022.9760932

Alturayeif, N., Alturaief, N., & Alhathloul, Z. (2020). DeepScratch: Scratch programming language extension for deep learning education. *International Journal of Advanced Computer Science and Applications, 11*(7). https://doi.org/10.14569/IJACSA.2020.0110777

Andersen, R., Mørch, A. I., & Litherland, K. T. (2022). Collaborative learning with block-based programming: Investigating human-centered artificial intelligence in education. *Behaviour & Information Technology*, *41*(9), 1830–1847. https://doi.org/10.1080/0144929x.2022.2083981

Baldoni, M., Baroglio, C., Bucciarelli, M., Capecchi, S., Gandolfi, E., Gena, C., Ianì, F., Marengo, E., Micalizio, R., Rapp, A., & Ras, I. N. (2024). Does any AI-based activity contribute to develop AI conception? A case study with Italian fifth and sixth grade classes. In *Proceedings of the AAAI Conference on Artificial Intelligence* (*Vol. 38, No. 21*, pp. 23060–23068). AAAI Press. https://doi.org/10.1609/aaai.v38i21.30350

Basu, S. (2019). Using rubrics integrating design and coding to assess middle school students' open-ended block-based programming projects. In *Proceedings of the 50th ACM Technical Symposium on Computer Science Education* (pp. 372–378). ACM. https://doi.org/10.1145/3287324.3287412

Broll, B., & Grover, S. (2023). Beyond black-boxes: Teaching complex machine learning ideas through scaffolded interactive activities. In *Proceedings of the AAAI Conference on Artificial Intelligence* (*Vol. 37, No. 13*, pp. 15990–15998). AAAI Press. https://doi.org/10.1609/aaai.v37i13.26898

Carrisi, M. C., Marras, M., & Vergallo, S. (2025). *A structured unplugged approach for foundational AI literacy in primary education* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2505.21398

Casal-Otero, L., Català, A., Fernández-Morante, C., Taboada, M., Cebreiro, B., & Barro, S. (2023). AI literacy in K–12: A systematic literature review. *International Journal of STEM Education, 10*(1), Article 29. https://doi.org/10.1186/s40594-023-00418-7

Chiu, T. K. F., Meng, H., Chai, C.-S., King, I., Wong, S., & Yam, Y. (2021). Creation and evaluation of a pre-tertiary artificial intelligence (AI) curriculum. *IEEE Transactions on Education*, *65*(1), 30–39. https://doi.org/10.48550/arXiv.2101.07570

Druga, S., & Ko, A. J. (2021). How do children's perceptions of machine intelligence change when training and coding smart programs? In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference* (pp. 416–427). ACM. https://doi.org/10.1145/3459990.3460712

Druga, S., Vu, S. T., Likhith, E., & Qiu, T. (2019). Inclusive AI literacy for kids around the world. In *Proceedings of FabLearn 2019: The 8th Annual Conference on Creativity and Fabrication in Education* (pp. 1–8). ACM. https://doi.org/10.1145/3311890.3311904

Emerson, A., Geden, M., Smith, A., Wiebe, E., Mott, B., Boyer, K. E., & Lester, J. (2020). Predictive student modeling in block-based programming environments with Bayesian hierarchical models. In *Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization* (pp. 280–289). ACM. https://doi.org/10.1145/3340631.3394853

Estevez, J., Garate, G., & Grana, M. (2019). Gentle introduction to artificial intelligence for high-school students using Scratch. *IEEE Access*, *7*, 179027–179036. https://doi.org/10.1109/access.2019.2956136

García, J. D. R., León, J. M., González, M. R., & Robles, G. (2019). Developing computational thinking at school with machine learning: An exploration. In *2019 International Symposium on Computers in Education (SIIE)* (pp. 1–6). IEEE. https://doi.org/10.1109/SIIE48397.2019.8970124

Grover, S. (2024). Teaching AI to K-12 learners: Lessons, issues, and guidance. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (pp. 20–21). ACM. https://doi.org/10.1145/3626252.3630937

Grover, S., Broll, B., & Babb, D. (2023). Cybersecurity education in the age of AI: Integrating AI learning into cybersecurity high school curricula. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (pp. 121–127). ACM. https://doi.org/10.1145/3545945.3569750

Hsu, T.-C., Abelson, H., Lao, N., Tseng, Y.-H., & Lin, Y.-T. (2021). Behavioral-pattern exploration and development of an instructional tool for young children to learn AI. *Computers and Education: Artificial Intelligence*, *2*, Article 100012. https://doi.org/10.1016/j.caeai.2021.100012

Hsu, T.-C., Abelson, H., & Van Brummelen, J. (2022). The effects on secondary school students of applying experiential learning to the conversational AI learning curriculum. *The International Review of Research in Open and Distributed Learning, 23*(1), 82–103. https://doi.org/10.19173/irrodl.v22i4.5474

Irgens, G. A., Vega, H., Adisa, I., & Bailey, C. (2022). *Characterizing children's conceptual knowledge and computational practices in a critical machine learning educational programme*. *International Journal of Child-Computer Interaction, 34*, Article 100541. https://doi.org/10.1016/j.ijcci.2022.100541

Jayasuriya, S., Swisher, K., Rego, J. D., Chandran, S., Mativo, J., Kurz, T., Collins, C. E., Robinson, D. T., & Pidaparti, R. (2024). ImageSTEAM: Teacher professional development for integrating visual computing into middle school lessons. *Proceedings of the AAAI Conference on Artificial Intelligence, 38*(21), 23101–23109. https://doi.org/10.1609/aaai.v38i21.30355

Kandlhofer, M., Steinbauer, G., Hirschmugl-Gaisch, S., & Huber, P. (2016). Artificial intelligence and computer science in education: From kindergarten to university. In *2016 IEEE Frontiers in Education Conference (FIE)* (pp. 1–9). IEEE. https://doi.org/10.1109/FIE.2016.7757353

Kaplan, R., & Meylani, R. (2025). Dimensions of artificial intelligence literacy: A qualitative synthesis of contemporary research literature. *Journal of Computer and Education Research, 13*(26), 790–812. https://doi.org/10.18009/jcer.1648380

Kaspersen, M. H., Bilstrup, K.-E. K., Van Mechelen, M., Hjorth, A., Bouvin, N. O., & Petersen, M. G. (2021). VotestratesML: A high school learning tool for exploring machine learning and its societal implications. In *Proceedings of FabLearn Europe / MakeEd 2021 - An International Conference on Computing, Design and Making in Education* (pp. 1–10). ACM. https://doi.org/10.1016/j.ijcci.2022.100539

Kazemitabaar, M., Chow, J., Ma, C. K. T., Ericson, B. J., Weintrop, D., & Grossman, T. (2023). Studying the effect of AI code generators on supporting novice learners in introductory programming. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (pp. 1–16). ACM. https://doi.org/10.48550/arXiv.2302.07427

Kim, K., & Kwon, K. (2024). Tangible computing tools in AI education: Approach to improve elementary students' knowledge, perception, and behavioral intention towards AI. *Education and Information Technologies*. https://doi.org/10.1007/s10639-024-12497-2

Kong, S.-C., Cheung, W. M.-Y., & Tsang, O. (2022). Evaluating an artificial intelligence literacy programme for empowering and developing concepts, literacy and ethical awareness in senior secondary students. *Education and Information Technologies*, *28*(4), 4703–4724. https://doi.org/10.1007/s10639-022-11408-7

Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, *33*(1), 159–174. https://doi.org/10.2307/2529310

Long, D., & Magerko, B. (2020). What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). ACM. https://doi.org/10.1145/3313831.3376727

Long, D., Teachey, A., & Magerko, B. (2022). Family learning talk in AI literacy learning activities. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). ACM. https://doi.org/10.1145/3491102.3502091

McCarthy, J. (1987). Generality in artificial intelligence. *Communications of the ACM*, *30*(12), 1029–1034. https://doi.org/10.1145/33447.33448

Memari, M., & Ruggles, K. (2025). *Artificial intelligence in elementary STEM education: A systematic review of current applications and future challenges* (arXiv:2511.00105). arXiv. https://doi.org/10.48550/arXiv.2511.00105

MoNE. (2023). *FEYZA projesi* [Republic of Türkiye, Ministry of National Education, FEYZA Project]. https://dogmprojeler.meb.gov.tr/www/feyza-projesi/icerik/4

Moreno-León, J., Vasco-González, M., Román-González, M., & Robles, G. (2024). Investigating the impact of programming activities on computational thinking and AI literacy in Spanish schools. In *Proceedings of the 19th WiPSCE Conference on Primary and Secondary Computing Education Research* (pp. 1–10). ACM. https://doi.org/10.1145/3677619.3678111

Ng, D. T. K., Su, J., & Chu, S. K. W. (2023). Fostering secondary school students' AI literacy through making AI-driven recycling bins. *Education and Information Technologies*, *29*(8), 9715–9746. https://doi.org/10.1007/s10639-023-12183-9

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, J. K., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, *372*, Article n71. https://doi.org/10.1136/bmj.n71

Park, Y., & Shin, Y. (2021). Tooee: A novel Scratch extension for K-12 big data and artificial intelligence education using text-based visual blocks. *IEEE Access*, *9*, 149630–149646. https://doi.org/10.1109/access.2021.3125060

Park, Y., & Shin, Y. (2022a). A block-based interactive programming environment for large-scale machine learning education. *Applied Sciences*, *12*(24), Article 13008. https://doi.org/10.3390/app122413008

Park, Y., & Shin, Y. (2022b). *Text processing education using a block-based programming language*. *IEEE Access*, *10*, 128484–128497. https://doi.org/10.1109/ACCESS.2022.3227765

Percival, N., Rayavaram, P., Narain, S., & Lee, C. S. (2022). CryptoScratch: Developing and evaluating a block-based programming tool for teaching K-12 cryptography education using Scratch. In *2022 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1–6). IEEE. https://doi.org/10.48550/arXiv.2302.11606

Sabuncuoglu, A. (2020). Designing one year curriculum to teach artificial intelligence for middle school. In *Proceedings of the 2020 ACM Conference on Innovation and Technology in Computer Science Education* (pp. 248–254). ACM. https://doi.org/10.1145/3341525.3387364

Sanusi, I. T., Oyelere, S. S., Vartiainen, H., Suhonen, J., & Tukiainen, M. (2023). Developing middle school students' understanding of machine learning in an African school. *Computers and Education: Artificial Intelligence*, *5*, Article 100155. https://doi.org/10.1016/j.caeai.2023.100155

Schaper, M.-M., Tamashiro, M. A., Smith, R. C., Van Mechelen, M., & Iversen, O. S. (2023). Five design recommendations for teaching teenagers' about artificial intelligence and machine learning. In *Proceedings of the 22nd Annual ACM Interaction Design and Children Conference* (pp. 13–24). ACM. https://doi.org/10.1145/3585088.3589366

Su, J., & Yang, W. (2023). AI literacy curriculum and its relation to children's perceptions of robots and attitudes towards engineering and science: An intervention study in early childhood education. *Journal of Computer Assisted Learning*, *40*(1), 241–253. https://doi.org/10.1111/jcal.12867

Su, J., & Zhong, Y. (2022). Artificial intelligence (AI) in early childhood education: Curriculum design and future directions. *Computers and Education: Artificial Intelligence*, *3*, Article 100072. https://doi.org/10.1016/j.caeai.2022.100072

Tedre, M., Toivonen, T., Kahila, J., Vartiainen, H., Valtonen, T., Jormanainen, I., & Pears, A. (2021). Teaching machine learning in K–12 classroom: Pedagogical and technological trajectories for artificial intelligence education. *IEEE Access*, *9*, 110558–110572. https://doi.org/10.1109/access.2021.3097962

Touretzky, D., Gardner-McCune, C., Martin, F., & Seehorn, D. (2019). Envisioning AI for K-12: What should every child know about AI? In *Proceedings of the AAAI Conference on Artificial Intelligence* (*Vol. 33, No. 01*, pp. 9795–9799). AAAI Press. https://doi.org/10.1609/aaai.v33i01.33019795

UNESCO. (2019). *International conference on artificial intelligence and education, planning education in the AI era: Lead the leap: Final report.* UNESCO.

Van Brummelen, J., Heng, T., & Tabunshchyk, V. (2021). Teaching tech to talk: K-12 conversational artificial intelligence literacy curriculum and development tools. In *Proceedings of the AAAI Conference on Artificial Intelligence* (*Vol. 35, No. 17*, pp. 15655–15663). AAAI Press. https://doi.org/10.1609/aaai.v35i17.17844

Vartiainen, H., Tedre, M., & Valtonen, T. (2020). Learning machine learning with very young children: Who is teaching whom? *International Journal of Child-Computer Interaction*, *25*, Article 100182. https://doi.org/10.1016/j.ijcci.2020.100182

Vincent-Lancrin, S., & Vlies, R. v. d. (2020). *Trustworthy artificial intelligence (AI) in education*. OECD Publishing. https://doi.org/10.1787/a6c90fa9-en

Von Wangenheim, G. C., Hauck, J. C. R., Pacheco, F. S., & Bertonceli Bueno, M. F. (2021). Visual tools for teaching machine learning in K-12: A ten-year systematic mapping. *Education and Information Technologies*, *26*(5), 5733–5778. https://doi.org/10.1007/s10639-021-10570-8

Williams, R., Ali, S., Alcantara, R., Burghleh, T., Alghowinem, S., & Breazeal, C. (2024). Doodlebot: An educational robot for creativity and AI literacy. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 772–780). ACM. https://doi.org/10.1145/3610977.3634950

Williams, R., Kaputsos, S. P., & Breazeal, C. (2021). Teacher perspectives on how to train your robot: A middle school AI and ethics curriculum. *Proceedings of the AAAI Conference on Artificial Intelligence, 35*(17), 15678–15686. https://doi.org/10.1609/aaai.v35i17.17847

Williams, R., Park, H. W., & Breazeal, C. (2019). A is for artificial intelligence. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Paper 227, pp. 1–11). ACM. https://doi.org/10.1145/3290605.3300677

Yim, I. H. Y., & Su, J. (2025). Artificial intelligence literacy education in primary schools: A review. *International Journal of Technology and Design Education, 35*, 2175–2204. https://doi.org/10.1007/s10798-025-09979-w

Yoder, S., Tatar, C., Aderemi, I., Boorugu, S., Jiang, S., & Akram, B. (2021). Gaining insight into effective teaching of AI problem-solving through CSEDM: A case study (work in progress). In *Proceedings of the 14th International Conference on Educational Data Mining* (pp. 671–674). International Educational Data Mining Society.

Zhou, Z., Jin, J., Phadnis, V., Yuan, X., Jiang, J., Qian, X., He, J., & Du, R. (2024). Experiencing InstructPipe: Building multi-modal AI pipelines via prompting LLMs and visual programming. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1–8). ACM. https://doi.org/10.1145/3613905.3648656